

Disproof of conjecture 00000000238: D_7 is not the densest lattice packing in dimension 7

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: The sphere packing density is the supremal fraction of space filled by non-overlapping unit balls. Conjecture: The densest lattice packings in dimensions five, six, and seven are given by the D lattices; and the determination is settled through Voronoi cell volumes. (D-lattice Voronoi volume optimality in dimensions 5-7) (The Chinese text states the same.)

Reading. The main clause is a conjunction over $n \in \{5, 6, 7\}$ of

(C_n) the densest lattice packing of unit balls in $\mathbb{R}^n$ is given by D_n ,

where $D_n = \{x \in \mathbb{Z}^n : x_1 + \dots + x_n \text{ even}\}$ is the standard root lattice. We refute (C₇), which refutes the conjunction and hence the conjecture. We read “given by D_7 ” as: the unit balls are centred at the points of a similar copy $c\varphi(D_7)$ of D_7 (any scale $c \neq 0$, any linear isometry φ). We refute two readings:

- (R1) some lattice packing of unit balls centred at a similar copy of D_7 is at least as dense as every lattice packing of unit balls (`not_D7_densest`);
- (R2) (stronger) there is a lattice packing of unit balls that is *strictly* denser than every lattice packing of unit balls centred at a similar copy of D_7 (`exists_packing_denser_than_D7`). So no maximizer is of D_7 type.

Not addressed. The cases $n = 5$ and $n = 6$ are not used. The trailing clause “the determination is settled through Voronoi cell volumes” describes a method; it is the second part of a conjunction and is not needed. (The Voronoi cell of a lattice is a fundamental domain, so its volume is the covolume used below.)

2 Definitions (as in the Lean file)

We work in $\mathbb{R}^n = \text{EuclideanSpace } \mathbb{R} \text{ (Fin } n\text{)}$ with Lebesgue measure `vol` (Mathlib’s `volume`) and Euclidean distance.

Definition 1 (lattice packing of unit balls; `LatticePacking n`). A $\mathbb{Z}$ -submodule $\Lambda \subset \mathbb{R}^n$ that is discrete and spans $\mathbb{R}^n$ over $\mathbb{R}$ (Mathlib’s `DiscreteTopology` and `IsZLattice`), such that for distinct $x, y \in \Lambda$ the open unit balls $B(x, 1)$ and $B(y, 1)$ are disjoint. Equivalently $|x - y| \geq 2$ for $x \neq y$ in Λ (Mathlib `disjoint_ball_ball_iff`); compare the Leech lattice article: “The last condition is equivalent to the condition that unit balls centered at the points of Λ_{24} do not overlap.”

Definition 2 (density; `LatticePacking.density`). $\Delta(P) = \text{vol}(B(0, 1)) / \text{covol}(\Lambda)$, where `covol` is Mathlib’s `ZLattice.covolume`, the volume of a fundamental domain (“The n-dimensional volume of this fundamental domain is sometimes called the covolume of the lattice”, Wikipedia, Lattice (group)).

Lemma 3 (`LatticePacking.filled_fraction`). *For every fundamental domain F of Λ , $\text{vol}((\bigcup_{x \in \Lambda} B(x, 1)) \cap F) / \text{vol}(F) = \Delta(P)$.*

Proof. $\text{vol}(B(0, 1)) = \sum_{g \in \Lambda} \text{vol}((g + B(0, 1)) \cap F)$ (Mathlib `IsAddFundamentalDomain.measure_eq_tsum`), $g + B(0, 1) = B(g, 1)$, and the sets $B(g, 1) \cap F$ are pairwise disjoint, so the sum is $\text{vol}((\bigcup_g B(g, 1)) \cap F)$. Finally $\text{covol}(\Lambda) = \text{vol}(F)$. $\square$

So $\Delta(P)$ is the proportion of each fundamental cell, hence (by periodicity) of space, that the balls fill; “The proportion of space filled by the spheres is called the packing density of the arrangement” (Wikipedia, Sphere packing).

Definition 4 (`Dlat n`, `IsSimilarCopy`, `DLatticeIsDensest`). $D_n = \{x \in \mathbb{R}^n : \exists z \in \mathbb{Z}^n, x = z, \sum_i z_i \text{ even}\}$ (“The D_n root lattice ... consists of all integer vectors whose components sum to an even integer”, Wikipedia, Root system). Λ is a similar copy of M if $\Lambda = \{c\varphi(x) : x \in M\}$ for some real $c \neq 0$ and linear isometry φ of $\mathbb{R}^n$. `DLatticeIsDensest n` says: there is a lattice packing Q with Λ_Q a similar copy of D_n and $\Delta(P) \leq \Delta(Q)$ for every lattice packing P .

3 Proof

Lemma 5 (`covolume_span`). *If b is an $\mathbb{R}$ -basis of $\mathbb{R}^7$ then $\text{covol}(\text{span}_{\mathbb{Z}} b) = |\det b|$ (determinant in the standard basis).*

Proof. Mathlib `ZLattice.covolume_eq_det_mul_measureReal` with the standard basis, whose fundamental parallelepiped (the unit cube) has volume 1. $\square$

The witness (Construction A). Let $C \subset \mathbb{F}_2^7$ be the binary $[7, 3, 4]$ simplex code with generators $g_1 = 1000111$, $g_2 = 0101011$, $g_3 = 0011101$. Its seven nonzero codewords all have weight 4; Lean checks $\text{weight} \geq 4$ by `decide` over the eight messages (`weight_ge`). Let $E = \{x \in \mathbb{Z}^7 : x \bmod 2 \in C\}$ (`E7A`).

Lemma 6 (`sum_sq_ge`, `E7A_norm`). *Every nonzero $x \in E$ has $|x| \geq 2$.*

Proof. If $x \bmod 2 = 0$, some coordinate is a nonzero even integer, so $|x|^2 \geq 4$. Otherwise $x \bmod 2$ is a nonzero codeword, so at least 4 coordinates are odd, and $|x|^2 \geq 4$. $\square$

Let Λ_0 be the $\mathbb{Z}$ -span of the rows of $A_L = \text{rows } g_1, g_2, g_3, 2e_4, 2e_5, 2e_6, 2e_7$ (as integer vectors). The rows lie in E , so $\Lambda_0 \subseteq E$ (`span_1e_E7A`). The matrix A_L is upper triangular with diagonal $1, 1, 1, 2, 2, 2, 2$, so $\det A_L = 16$ (`det_AL`). Hence $P_0 = (\Lambda_0)$ is a lattice packing of unit balls (`PE7`) with $\Delta(P_0) = \text{vol}(B(0, 1)) / 16$ (`PE7_density`).

Remark 7. In fact $\Lambda_0 = E$ (both have index 16 in $\mathbb{Z}^7$), and E is an integral model of $\sqrt{2} E_7$. Neither fact is used or formalized.

Lemma 8 (`Dlat_eq_span`, `det_AD`). *D_7 is the $\mathbb{Z}$ -span of $e_j + e_7$ ($1 \leq j \leq 6$) and $2e_7$; this basis matrix A_D is upper triangular with $\det A_D = 2$.*

Proof. The generators have even coordinate sum. Conversely, if $\sum_i z_i = 2k$ then $z = \sum_{j \leq 6} z_j(e_j + e_7) + (k - \sum_{j \leq 6} z_j) 2e_7$. $\square$

Theorem 9 (`similar_D7_density_lt`). *If Q is a lattice packing of unit balls with $\Lambda_Q = c\varphi(D_7)$, then $\Delta(Q) < \Delta(P_0)$.*

Proof. Λ_Q is the $\mathbb{Z}$ -span of the basis $c\varphi(\text{rows of } A_D)$, so $\text{covol}(\Lambda_Q) = |c|^7 |\det \varphi| \cdot 2 = 2|c|^7$, using $|\det \varphi| = 1$ (`abs_det_isometry`, from Mathlib’s `OrthonormalBasis.det_to_matrix_orthonormalBasis_real`). The vector $v = e_1 + e_7 \in D_7$ has $|v| = \sqrt{2}$, so $c\varphi(v) \in \Lambda_Q$ is nonzero of length $|c|\sqrt{2}$. Non-overlap of the balls at $c\varphi(v)$ and 0 gives $|c|\sqrt{2} \geq 2$, so $|c| \geq \sqrt{2}$ and $|c|^7 \geq 8\sqrt{2}$. Therefore $\Delta(Q) = \text{vol}(B(0,1))/(2|c|^7) \leq \text{vol}(B(0,1))/(16\sqrt{2}) < \text{vol}(B(0,1))/16 = \Delta(P_0)$. $\square$

Taking $P = P_0$ gives (R2), and (R2) implies (R1) is false. In center-density terms: P_0 has $1/16$ and the best D_7 packing ($c = \sqrt{2}$) has $2^{-9/2} \approx 0.0442$.

4 Lean formalization and axiom audit

File `lean/Conjecture238/Basic.lean` (360 lines, only `import Mathlib`), namespace `C238`. Main theorems: (Unicode symbols of the Lean source are transliterated to ASCII here.)

```
theorem exists_packing_denser_than_D7 :
  exists P : LatticePacking 7, forall Q : LatticePacking 7,
    IsSimilarCopy (Dlat 7) Q.Lambda -> Q.density < P.density
theorem not_D7_densest : not (DLatticeIsDensest 7)
theorem conjecture_238_false :
  not (forall n in ({5, 6, 7} : Finset Nat), DLatticeIsDensest n)
theorem LatticePacking.filled_fraction {n : Nat} (P : LatticePacking n)
  {F : Set (Rn n)} (hF : IsAddFundamentalDomain P.Lambda F volume) :
  volume.real ((Union x : P.Lambda, ball (x : Rn n) 1) inter F)
  / volume.real F = P.density
```

Axiom audit. `#print axioms` for each of these theorems reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide`, and no added axiom. `lake build` succeeds.

Conventions. Balls are open Euclidean balls. “Lattice” means a discrete, full-rank $\mathbb{Z}$ -submodule (Mathlib `IsZLattice`). Similar copies allow any real $c \neq 0$ and any linear isometry. Density is $\text{vol}(B(0,1))/\text{covol}$, which equals the filled fraction of any fundamental domain (Lemma `filled_fraction`). The volume of $B(0,1)$ is never computed; it cancels.